



Seasonal Agriculture Performance Analysis

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PROBLEM STATEMENT

Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description

- A data analytics study of 4,000 farm records covering 28 attributes.
- Examines crop, season, location, weather, inputs, production, water use and financial performance.
- Data preparation includes missing-value treatment and duplicate checking.
- Exploration combines statistical summaries with univariate, bivariate and multivariate visualizations.
- Season-wise and crop-wise comparisons are used to identify meaningful differences and practical insights.
- The final output is a descriptive, evidence-based assessment of seasonal agricultural performance.

WHO ARE THE END USERS?

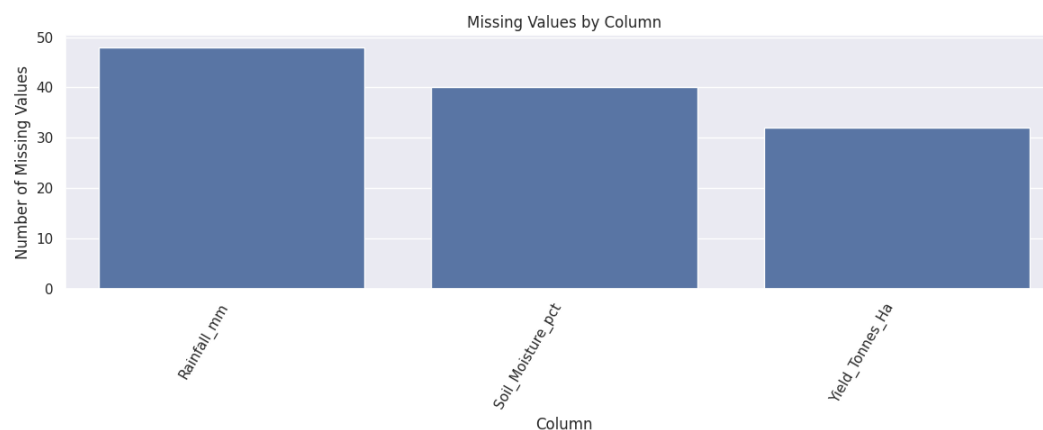
- Farmers and farm managers - to compare crop and seasonal performance.
- Agricultural planners - to support seasonal resource planning.
- Agricultural analysts and researchers - to study patterns in farm data.
- Policy and support teams - to identify areas that may need further investigation.

Technology Used

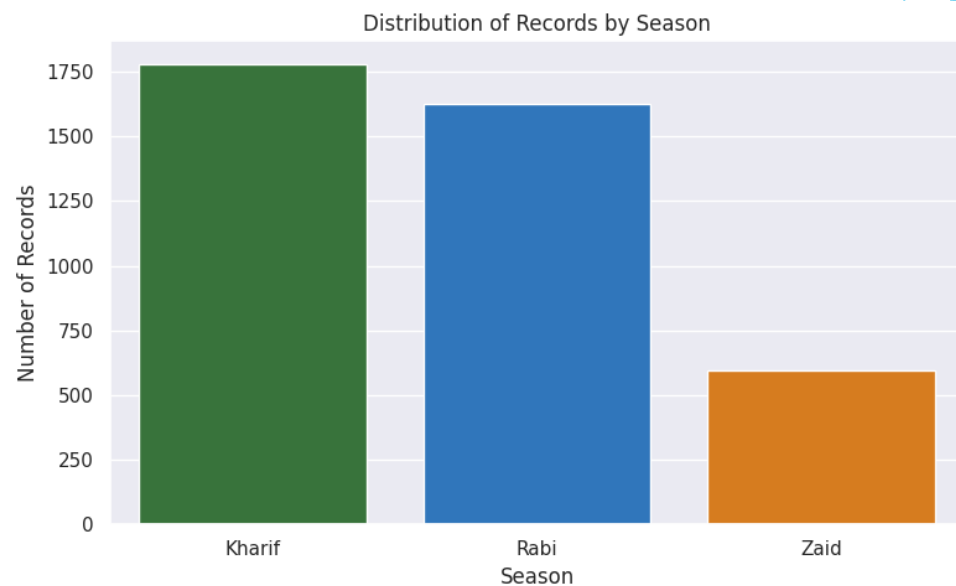
- Python - overall data analysis and visualization workflow.
- Pandas - data loading, cleaning, grouping and summaries.
- NumPy - numerical operations and statistical calculations.
- Matplotlib - customized analytical charts.
- Seaborn - statistical plots such as count plots, box plots and heatmaps.
- Google Colab - complete project documentation and execution environment.



RESULTS

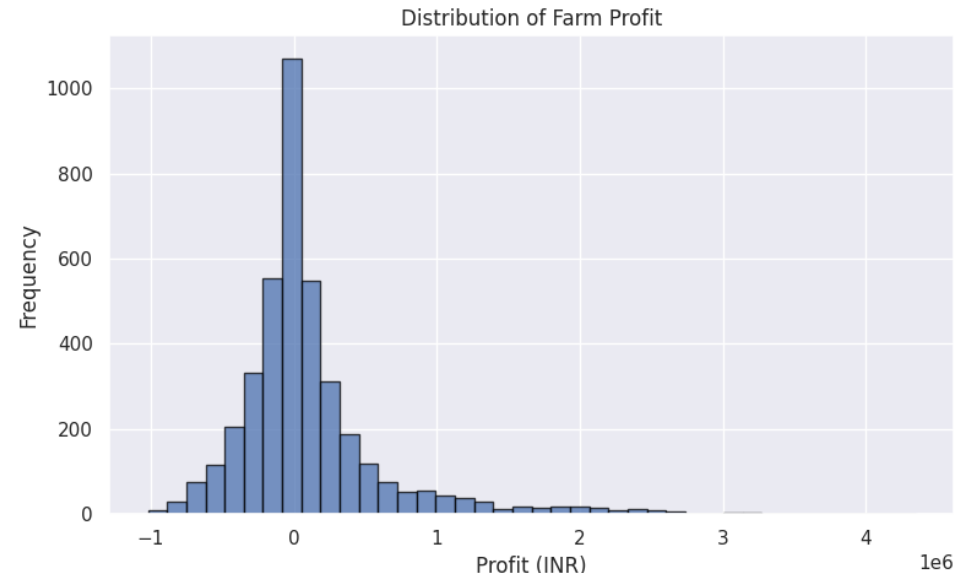
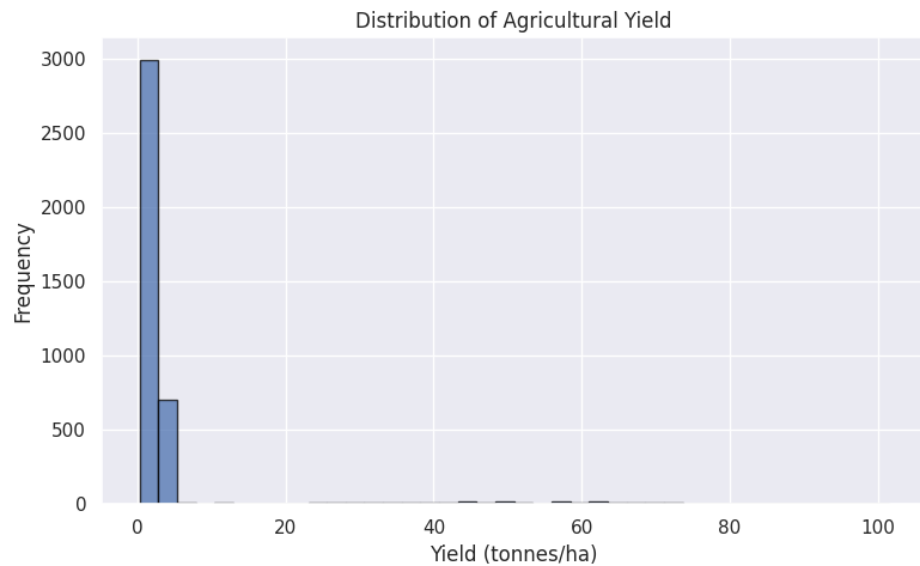


Missing values were limited to Rainfall (48), Soil Moisture (40) and Yield (32). Duplicate records were not found.



Season distribution: Kharif 1,779 records; Rabi 1,627; Zaid 594. Kharif is the largest group.

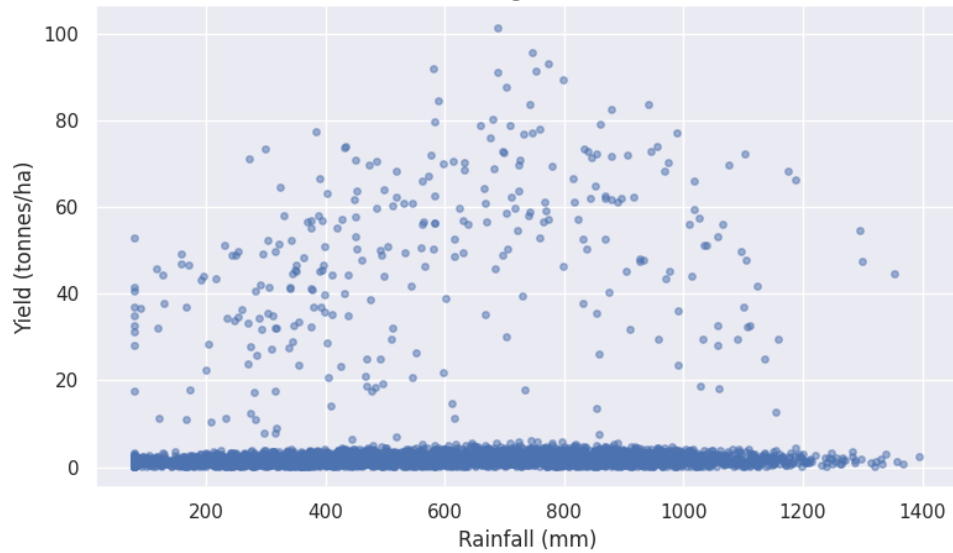
RESULTS



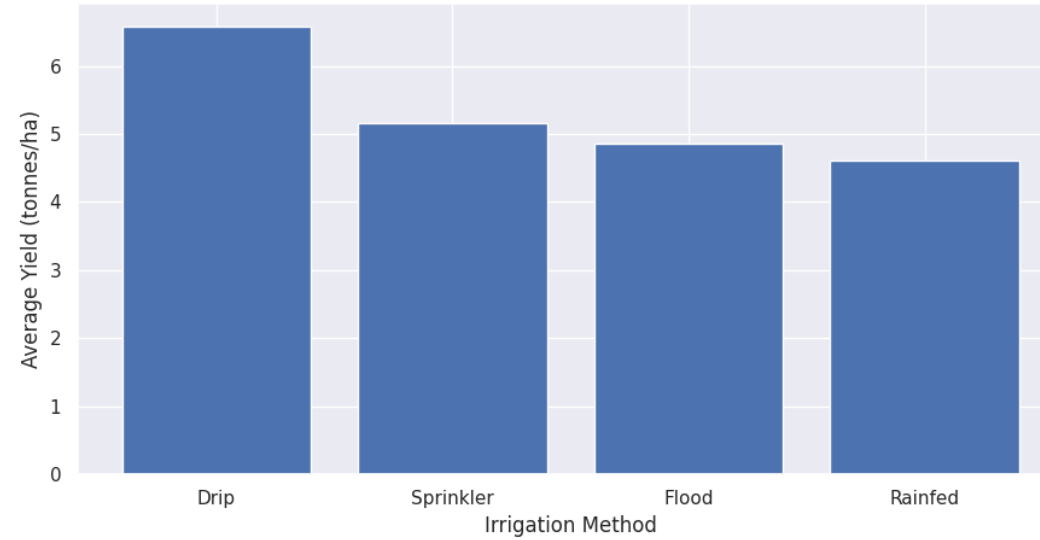
The dataset includes 8 crops. Yield is strongly right-skewed because a small number of observations have very high values. Profit also varies widely, showing both low/negative and high-return observations.

RESULTS

Rainfall vs Agricultural Yield

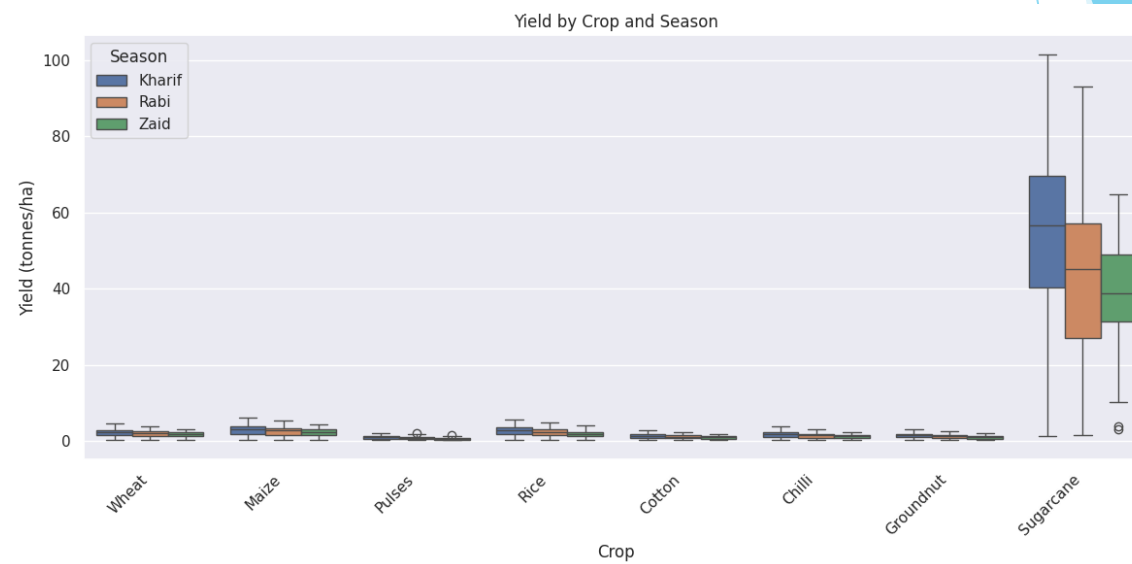
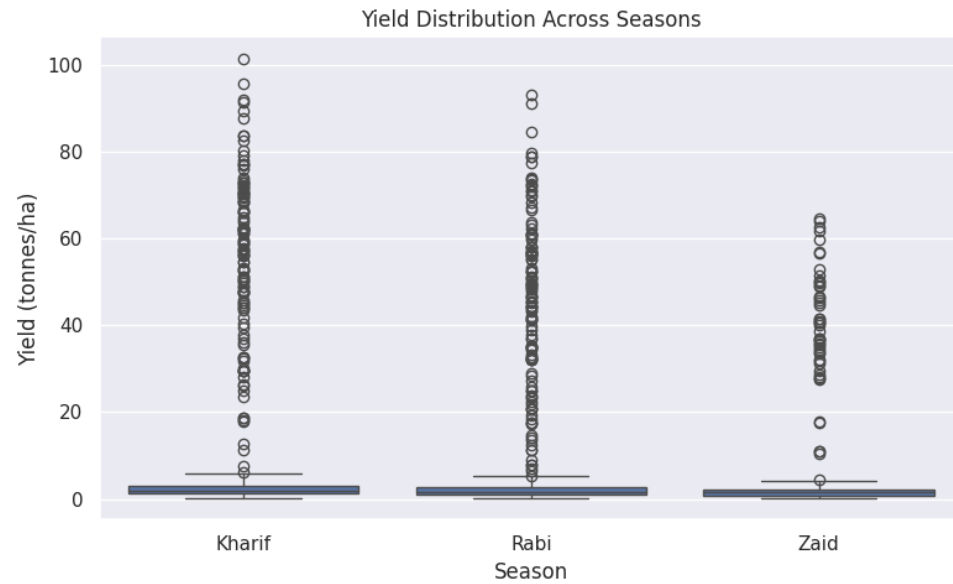


Average Yield by Irrigation Method



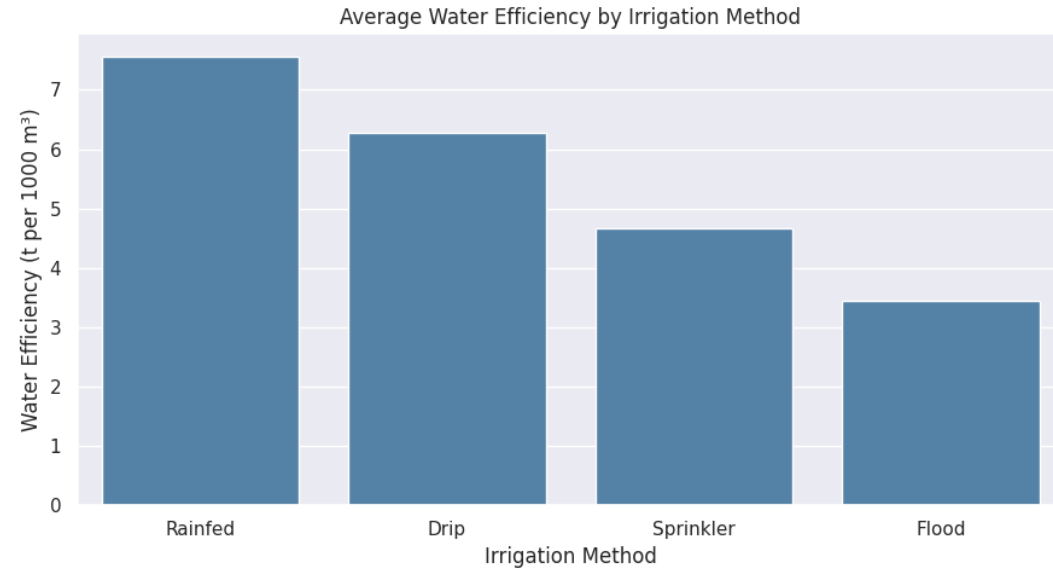
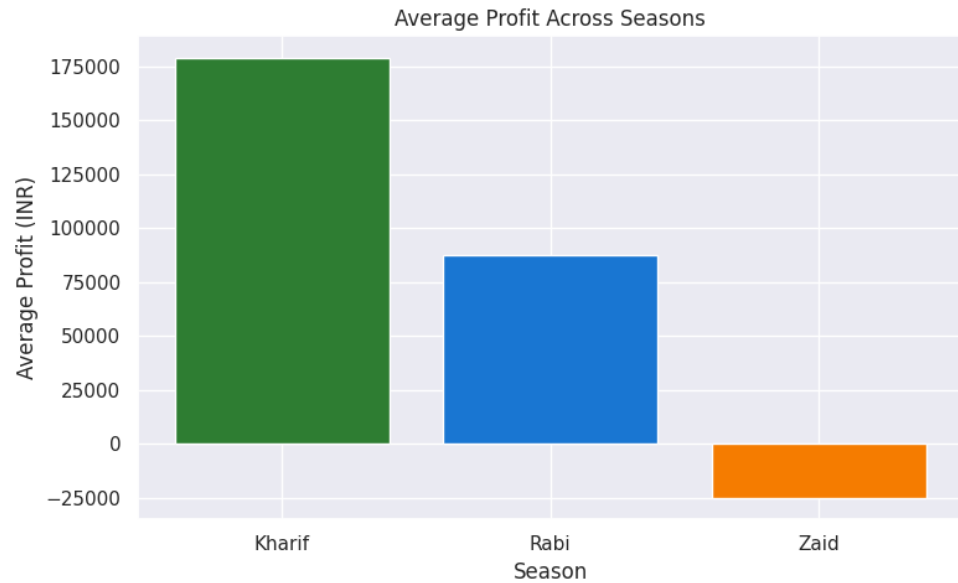
Rainfall and yield show a positive but scattered pattern. Larger farms generally produce more total output. Among irrigation methods, Drip has the highest average yield in this dataset.

RESULTS



Combining crop, season, rainfall, farm area, yield and profit gives a clearer picture of seasonal differences than examining single variables alone.

RESULTS



Season summary: Kharif — 5.63 t/ha average yield and ₹178,914.65 average profit; Rabi — 5.04 t/ha and ₹87,689.47; Zaid — 4.64 t/ha and -₹24,804.82. Rainfed has the highest average water efficiency, while Drip has the highest average yield.

Future scope

- Build an interactive dashboard for faster seasonal comparison.
- Extend the analysis with multi-year or real-time agricultural data.
- Add predictive models for yield, profit or seasonal risk when suitable data is available.
- Develop region-specific recommendations using finer-grained farm information.
- Combine weather forecasts with historical farm performance for planning support.

GitHub Link



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Thank you